

Week 7

Naive Bayes

*When Assumptions Meet
Motor Control Data*

Section	Question it answers
1. From Weights to Probabilities	What if we want confidence, not just a label?
2. Bayes' Theorem for Decoding	How do we invert encoding into decoding?
3. Testing NB on Our Data	Does it work on our four tasks?
4. Why NB Failed	What do synergies do to the independence assumption?
5. PCA to the Rescue	Can decorrelation fix what correlation broke?
6. Implications for Motor Control	What should researchers do about correlated features?
7. Bringing It Together	What did NB add to the comparison table?

1. From Weights to Probabilities

One muscle, one patient, one wrong intuition

A patient is referred to your rehabilitation clinic after a mild stroke. They perform a reaching task while you record EMG from their deltoid. The amplitude is unusually high. Your instinct says: elevated deltoid activation is a sign of impairment — this patient is probably motor-impaired. But is that right?

Here is what you know. In your clinic's referral stream, about 20% of patients turn out to have genuine motor impairment — the rest are referred for evaluation but ultimately test healthy. Among truly impaired patients, 80% show elevated deltoid activation (they over-recruit due to lost selectivity). But among healthy patients, 30% also show elevated deltoid activation — they simply reached faster or with more force. Given that this patient's deltoid is elevated, what is the probability they are actually impaired?

Most clinicians answer around 80%, reasoning from the test's accuracy. The correct answer is **40%**. The patient is more likely healthy than impaired, despite the elevated deltoid activation. This feels wrong — and understanding why it is right is the key to understanding Bayes' theorem.

Thinking it through with 100 patients

Of 100 patients referred to your clinic, 20 are truly impaired and 80 are healthy. Among the 20 impaired patients, 80% (= 16) show elevated deltoid activation. Among the 80 healthy patients, 30% (= 24) also show elevated deltoid activation. So of the 40 total patients with elevated deltoid activation, only 16 are truly impaired. That is 16 out of 40 = **40%**. The healthy patients outnumber the impaired patients so heavily that even a moderately accurate test produces more false alarms than true detections.

Bayes' theorem formalizes this reasoning into a single equation:

$$\begin{aligned} P(\text{impaired} \mid \text{elevated}) &= P(\text{elevated} \mid \text{impaired}) \cdot P(\text{impaired}) / P(\text{elevated}) \\ &= 0.80 \times 0.20 / (0.80 \times 0.20 + 0.30 \times 0.80) = 0.16 / 0.40 = \mathbf{0.40} \end{aligned}$$

Each term has a name and a role:

- **Prior** — $P(\text{impaired}) = 0.20$ — our belief before seeing the data. This is the base rate of impairment in the clinic. It is what most people forget to account for.
- **Likelihood** — $P(\text{elevated} \mid \text{impaired}) = 0.80$ — the probability of observing this evidence if the patient is truly impaired. This is the number our intuition fixates on.
- **Evidence** — $P(\text{elevated}) = 0.40$ — the total probability of seeing elevated deltoid activation across all patients. It normalizes the answer so the posteriors sum to one.
- **Posterior** — $P(\text{impaired} \mid \text{elevated}) = 0.40$ — our updated belief after seeing the data. The answer to the clinical question.

Bayes' theorem is a formal mechanism for updating beliefs with evidence. The posterior combines the likelihood (how consistent the evidence is with the hypothesis) with the prior (how plausible the hypothesis was to begin with). A strong likelihood (80%) gets pulled down by a low prior (20%) because there are so many more healthy patients generating false alarms.

From one muscle to six: the power of combining evidence

The deltoid alone gave us 40% — not enough to act on. But what if we also recorded the biceps, triceps, and three more muscles? Each muscle provides its own evidence about whether the patient is impaired. If the deltoid is elevated *and* the biceps shows abnormal timing *and* the triceps is under-recruited, the combined evidence might push the posterior much higher — or, if the other muscles look normal, pull it back down. This is exactly what **Naive Bayes** does: it applies Bayes' theorem with multiple features, combining each muscle's evidence into a single posterior probability.

NB is a **generative classifier** — instead of drawing a decision boundary directly (like logistic regression), it builds explicit profiles of what each class looks like. What firing rates or muscle amplitudes are typical for each direction, or for healthy and impaired subjects? Then it uses Bayes' theorem to compute the posterior probability of each class given the observed data. The output is a full probability distribution: for the 8-direction task, NB returns 8 probabilities summing to 1, telling us not only which direction was most likely but how much more likely it was than the runner-up.

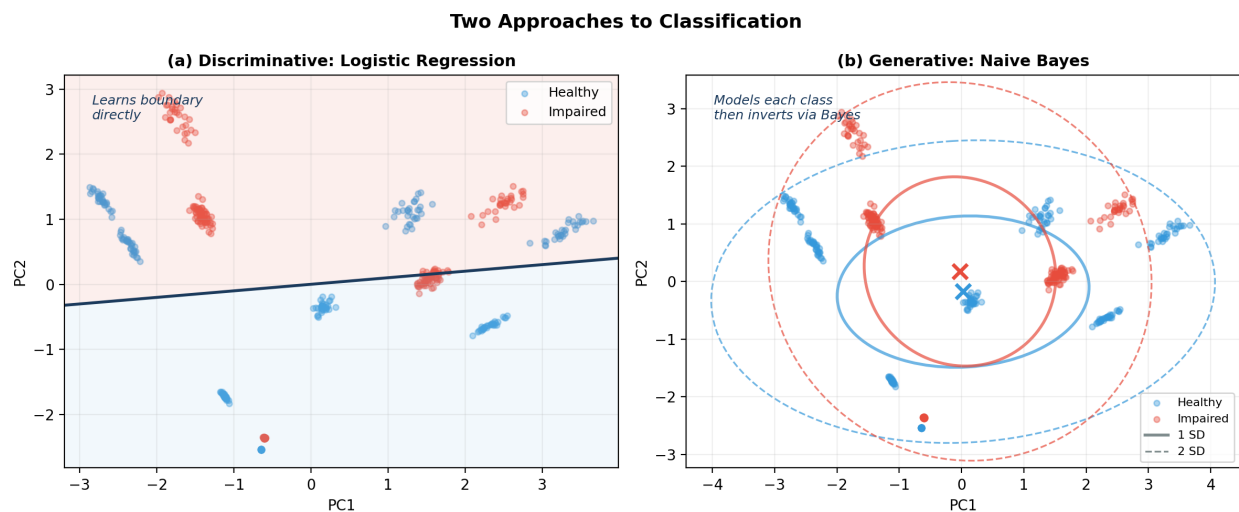


Figure 1. Two approaches to classification on our data (PCA space, binary task). (a) Logistic regression (discriminative): draws a decision boundary directly. (b) Naive Bayes (generative): fits a distribution to each class. Solid ellipses show 1 SD contours; dashed ellipses show 2 SD. A new point is assigned to whichever distribution gives it higher probability.

Where Weeks 5 and 6 left us

In Week 5, logistic regression established our baseline: 80.8% for direction decoding from EMG, 85.0% (AUC 0.94) for clinical diagnosis. LR gives us both a label and a probability through its sigmoid output — but the model has no understanding of what healthy or impaired patterns look like. It draws a boundary and assigns labels based on which side a trial falls.

In Week 6, we added 80 cosine-tuned neurons. The population vector decoded direction at 95.8% but gave no measure of uncertainty — no way to know if the model was sure or guessing between neighboring directions. Neural LR reached 91.2% for direction but only 73.3% for diagnosis (AUC 0.80), confirming the double dissociation.

Two gaps remain. First, the population vector gives a decoded direction but no confidence. Second, LR does not describe what each class looks like. NB fills both gaps: it models what each class looks like (generative) and outputs calibrated posteriors (Bayesian). But its independence

assumption — which makes it fast and simple — carries a cost we will discover on our correlated EMG features.

Bayes' theorem updates beliefs with evidence. The deltoid example showed that a single feature's evidence must be weighed against the prior — intuition alone is not enough. NB extends this to multiple features, combining each muscle or neuron's evidence into a posterior. Whether this works depends on whether the features provide independent evidence — the empirical question we test this week.

2. Bayes' Theorem for Decoding

Start with a concrete question. A subject reaches, 80 neurons fire, and we observe a vector of 80 firing rates. Which of the 8 target directions was the reach? Bayes' theorem tells us how to compute the answer:

$$P(\text{direction} = d \mid \text{rates}) = P(\text{rates} \mid \text{direction} = d) \cdot P(\text{direction} = d) / P(\text{rates})$$

Each term has a concrete motor control meaning. The **likelihood** $P(\text{rates} \mid d)$ asks: “if the reach was at 90° , how probable are these observed firing rates?” This is the encoding model from Week 6 run in reverse — we know what each neuron should fire for each direction, so we can evaluate how well the observed rates match each direction's expected profile. The **prior** $P(d)$ is our belief about each direction before seeing the data — uniform ($1/8$) in our balanced design, since each direction was equally likely. The **evidence** $P(\text{rates})$ is a normalization constant, the same for all directions, so we can ignore it when comparing.

The **posterior** $P(d \mid \text{rates})$ is the answer: a probability distribution over all 8 directions, given what we observed. The predicted direction is the one with the highest posterior — but the full distribution tells us more. A trial where one direction dominates ($P = 0.95$) is unambiguous. A trial where two neighboring directions are close ($P = 0.45$ vs $P = 0.40$) is the classifier saying “I am not sure — these two directions produce similar neural patterns.” This uncertainty is invisible to the population vector, which outputs a single angle regardless of confidence.

Worked example: one trial, five neurons

To make this concrete, consider Trial 8 from our dataset — a reach at 90° , decoded using five neurons with preferred directions spread around the circle. The observed firing rates are: Neuron 11 (PD $\approx 45^\circ$) = 8 spk/s, Neuron 21 (PD $\approx 90^\circ$) = 32 spk/s, Neuron 31 (PD $\approx 135^\circ$) = 19 spk/s, Neuron 41 (PD $\approx 180^\circ$) = 12 spk/s, Neuron 51 (PD $\approx 225^\circ$) = 10 spk/s.

NB asks: under each candidate direction, how likely are these rates? The key evidence comes from Neuron 21, whose preferred direction is 90° . It fired 32 spk/s — high, and close to what the model expects for 90° reaches but far above what it expects for 45° or 135° . Meanwhile, Neuron 51 (PD $\approx 225^\circ$) is quiet at 10 spk/s, which is consistent with 90° (opposite its preferred direction) but inconsistent with 135° or 180° . Each neuron casts an independent vote via its log-likelihood:

$$\begin{aligned} 90^\circ: -1.26 - 1.64 - 0.79 - 0.74 + 0.18 &= \mathbf{-4.26} & | & & 45^\circ: -1.67 - 3.14 - 0.49 - 0.61 \\ & - 0.23 = \mathbf{-6.14} & | & & 135^\circ: -0.25 - 3.49 - 1.33 - 1.61 - 1.80 = \mathbf{-8.48} \end{aligned}$$

How do log-likelihoods become probabilities? The differences between the sums are what matter. 90° beats 45° by $-4.26 - (-6.14) = 1.88$ log-units, and beats 135° by 4.22 log-units. To convert, we exponentiate these differences: $\exp(0) = 1.00$ for 90° , $\exp(-1.88) = 0.15$ for 45° , $\exp(-4.22) = 0.01$ for 135° (all other directions are negligibly close to zero). Then we normalize by dividing each by the sum ($1.00 + 0.15 + 0.01 = 1.17$):

$$\begin{aligned} P(90^\circ) &= 1.00 / 1.17 = \mathbf{0.86} & | & & P(45^\circ) &= 0.15 / 1.17 = \mathbf{0.13} & | & & P(135^\circ) &= 0.01 / 1.17 \\ & & & & & = \mathbf{0.01} \end{aligned}$$

The model is confident — 90° wins decisively — but the 13% assigned to 45° tells us exactly where the remaining uncertainty lies: the two neighboring directions that share Neuron 11's tuning curve. This is the kind of information a probability distribution provides that a single

decoded angle cannot.

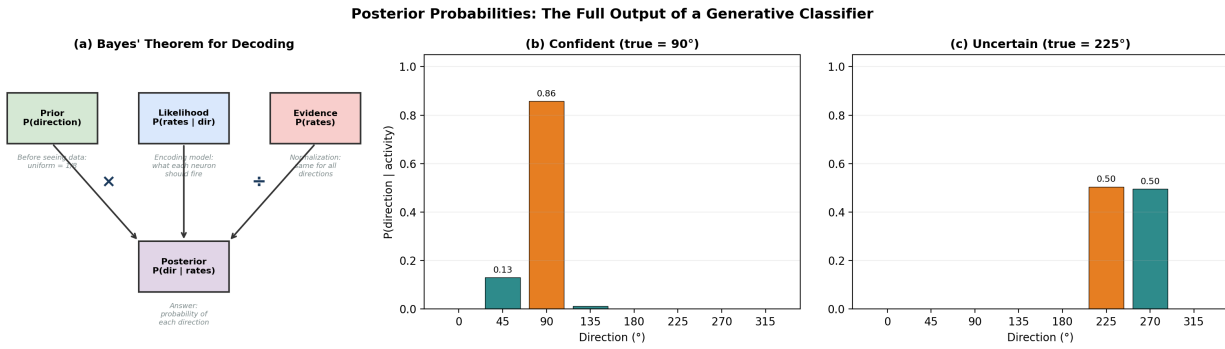


Figure 2. (a) Bayes' theorem for neural decoding: the prior (uniform over directions), likelihood (encoding model), and evidence (normalization) combine to produce the posterior. (b) A confident trial: the posterior concentrates on one direction (orange = true), with a small runner-up at the neighboring direction. (c) An uncertain trial: the posterior splits nearly evenly between two adjacent directions — the classifier is reporting that these movements produce similar neural patterns. Both panels use 5 neurons; with all 80, posteriors become near-certain for most trials.

3. Testing NB on Our Data

Section 2 showed NB decoding a single 90° reach correctly with five neurons. The posteriors were well-calibrated, the correct direction won, and the uncertainty landed on the neighboring direction. But that was one trial, hand-picked, with five neurons. The real test is whether this scales to the full dataset.

From one trial to 480

The procedure is the same as Weeks 5 and 6. We take all 480 trials (8 directions × 3 speeds × 20 subjects) and evaluate NB using leave-one-subject-out (LOSO) cross-validation: train on 19 subjects, test on the 20th, rotate through all 20. For each held-out trial, NB computes the posterior probability of each class — 8 posteriors for the direction task, 2 for the binary task — and we check whether the highest-posterior class matches the truth.

For direction decoding, NB now uses all 80 neurons (or all 6 muscles), not just five. It stores means and variances for every feature under every direction — 640 means and 640 variances for the neural task, 48 and 48 for the EMG task — and multiplies the likelihoods across all features, just as we did by hand for five neurons in Section 2. For the binary task, NB stores what each muscle or neuron looks like in healthy vs impaired subjects, then asks: does this patient's pattern better match the healthy profile or the impaired profile?

The calculation is identical to Section 2, just with more terms. For the 5-neuron example, we summed 5 log-likelihoods:

$$\log P(\text{rates} \mid 90^\circ) = \log P(r_{11} \mid 90^\circ) + \log P(r_{21} \mid 90^\circ) + \dots + \log P(r_{51} \mid 90^\circ) = \mathbf{-4.26} \quad (5 \text{ terms})$$

With 80 neurons, the same sum has 80 terms:

$$\log P(\text{rates} \mid 90^\circ) = \log P(r_1 \mid 90^\circ) + \log P(r_2 \mid 90^\circ) + \dots + \log P(r_{80} \mid 90^\circ) \quad (80 \text{ terms})$$

More terms means the differences between directions grow larger, making the posteriors more extreme — 80 neurons push most posteriors above 0.999, leaving almost no uncertainty. This is why NB with 80 neurons is so confident. But confidence is only useful if the model is correct. If features are redundant and the product rule double-counts their evidence, all that confidence points in the wrong direction.

We test four combinations — two tasks crossed with two feature sets. The two tasks are: (1) **direction decoding**, where the model sees one trial's recordings and predicts which of 8 reach directions the subject performed; and (2) **binary diagnosis**, where the model sees one trial and predicts whether the subject is healthy or motor-impaired. The two feature sets are: **6 EMG muscles** (surface recordings from the arm) and **80 M1 neurons** (simulated cortical recordings). This gives us four experiments: EMG × direction, EMG × diagnosis, neural × direction, and neural × diagnosis. The question is whether the posteriors that worked so well for 5 neurons and one trial hold up across the full dataset — and across both feature sets.

Direction decoding: NB ties or wins

On EMG features, NB achieves 81.2% — essentially identical to logistic regression's 80.8%. On neural features, NB reaches **95.2%**, outperforming logistic regression's 91.2% by 4 percentage points. NB on neural data nearly matches the population vector (95.8%) — both exploit the same

cosine tuning structure, one analytically, one probabilistically. For direction decoding, NB works as well as or better than everything we have tried.

Clinical diagnosis: the catastrophe

On EMG features, NB drops to **65.4%** — a 20 percentage point collapse from logistic regression's 85.0%, with AUC falling from 0.94 to 0.68. This is the worst performance on this task in the entire course — worse than chance-level on some folds. Yet on neural features, NB achieves 77.7%, actually *beating* logistic regression's 73.3% (AUC 0.86 vs 0.80).

Week 7 Results: Naive Bayes vs Logistic Regression

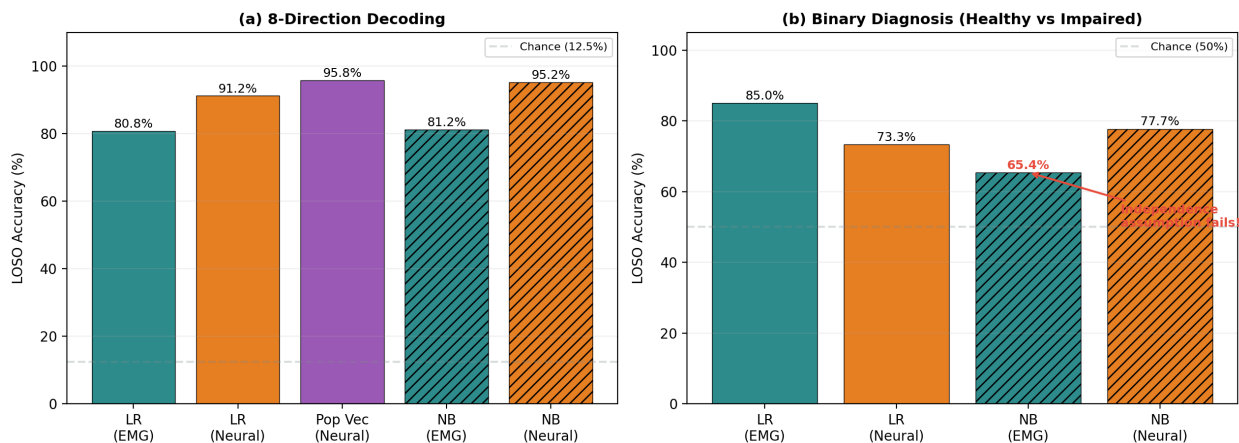


Figure 3. Naive Bayes vs logistic regression across both tasks and feature sets. Solid bars: LR (Weeks 5-6). Hatched bars: NB (Week 7). The 65.4% on EMG binary (red annotation) is a catastrophic failure. Yet NB wins on both neural tasks. Something about the EMG features is specifically toxic to NB.

ROC Curves: How Well Do Posterior Probabilities Separate Groups?

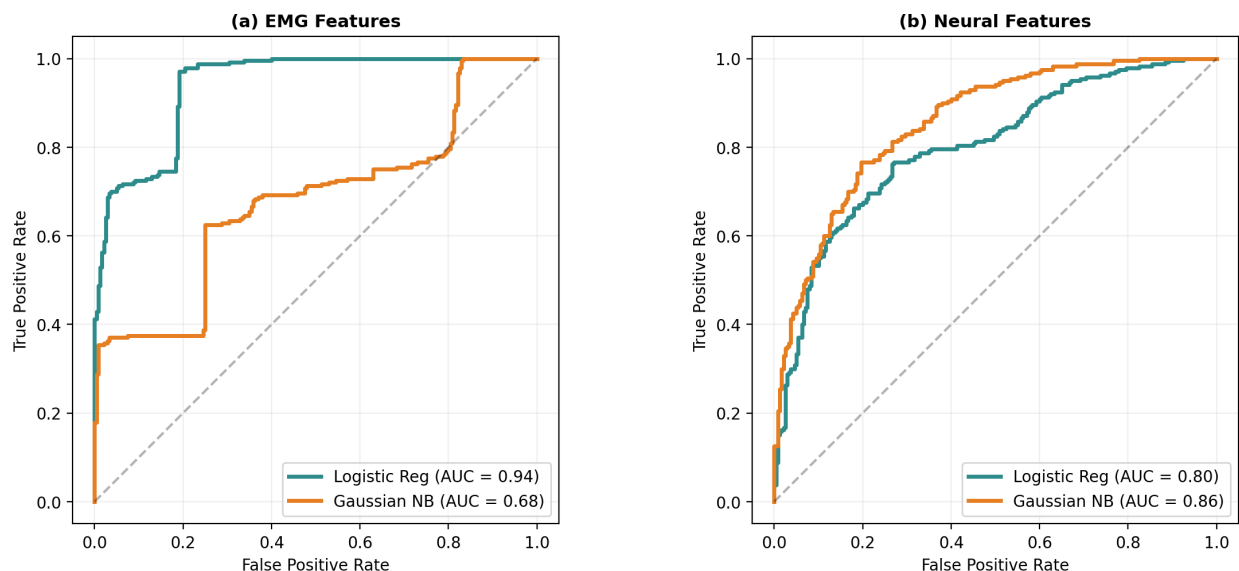


Figure 4. ROC curves for the binary task. (a) EMG: NB (AUC = 0.68) is dramatically worse than LR (AUC = 0.94) — the posterior probabilities are badly miscalibrated. (b) Neural: NB (AUC = 0.86) outperforms LR (AUC = 0.80). The difference between the two panels is the central puzzle of this week.

The pattern is striking. NB matches or beats LR on three of four tasks, then collapses by 20 percentage points on the fourth. This is not random noise — it is a systematic failure. Something

about EMG features combined with the subtle binary signal exposes a vulnerability in NB that direction decoding and neural features do not. What is it?

NB ties or beats LR on direction decoding and neural diagnosis. But on EMG binary classification, NB collapses to 65.4% — the worst result in the course. Three wins and a catastrophe.

4. Why NB Failed — Synergies and the Independence Assumption

Section 3 showed a clear pattern: NB collapsed on EMG binary (65.4%) but succeeded on the other three tasks. To understand why, we need to revisit what NB assumes and check whether that assumption holds for our two feature sets.

The product rule assumes independence

Recall from Section 2 that NB classifies by multiplying likelihoods across features. Each muscle or neuron contributes an independent term to the product. This assumes that, given the class label, knowing one feature's value tells you nothing about another's. Whether that assumption holds depends on how correlated the features actually are.

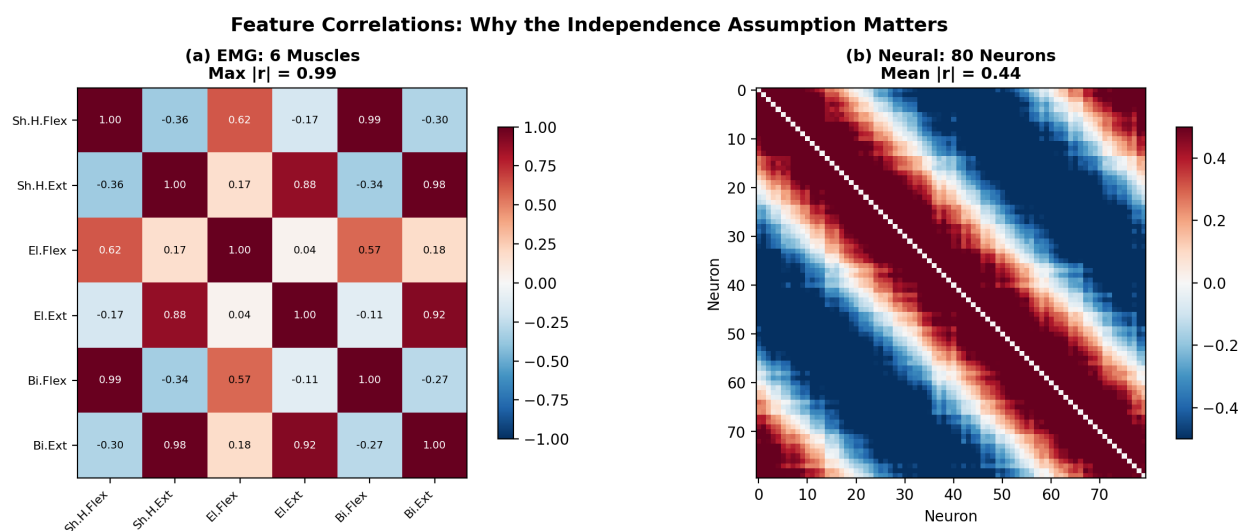


Figure 5. Feature correlations. (a) The 6×6 EMG correlation matrix: some muscle pairs co-activate with r up to 0.99. These are the motor synergies from Week 4 — muscles within the same synergy module fire together because the nervous system controls them as a unit. (b) The 80×80 neural correlation matrix: lower and more distributed correlations. Each neuron has a unique preferred direction, so their activity patterns are more diverse.

How correlations cause double-counting

The EMG correlation matrix reveals the problem. The shoulder flexor (Sh.H.Flex) and the bi-articular flexor (Bi.Flex) co-activate at $r = 0.99$ — both cross the shoulder joint and are recruited together during flexion movements. Similarly, the shoulder extensor and bi-articular extensor co-activate at $r = 0.98$. These are the synergy modules from Week 4: single-joint and bi-articular muscles of the same type fire together because the nervous system controls them as a unit. When NB evaluates these muscle pairs independently, it counts their evidence twice. If both flexors are elevated in a patient, NB treats that as two separate pieces of evidence for impairment, when it is really one piece of evidence (the flexor synergy is active) expressed in two measurements. The product rule has no way to recognize that these muscles are redundant. It multiplies their likelihoods as though each provides fresh information, inflating the posterior in whichever direction the redundant evidence points.

But this raises two questions. First: NB also used EMG for direction decoding and scored 81.2% — so why didn't the correlations destroy that task too? Second: NB succeeded on neural features for

both tasks — what makes neurons different from muscles?

Why neural features are better suited to NB

The neural correlation matrix (Figure 5b) tells a different story. Eighty neurons with diverse preferred directions have lower, more distributed correlations. To see why this helps NB, consider what each neuron contributes to the product rule. Figure 6 shows the class-conditional Gaussians for three example neurons — each one separates a different pair of directions:

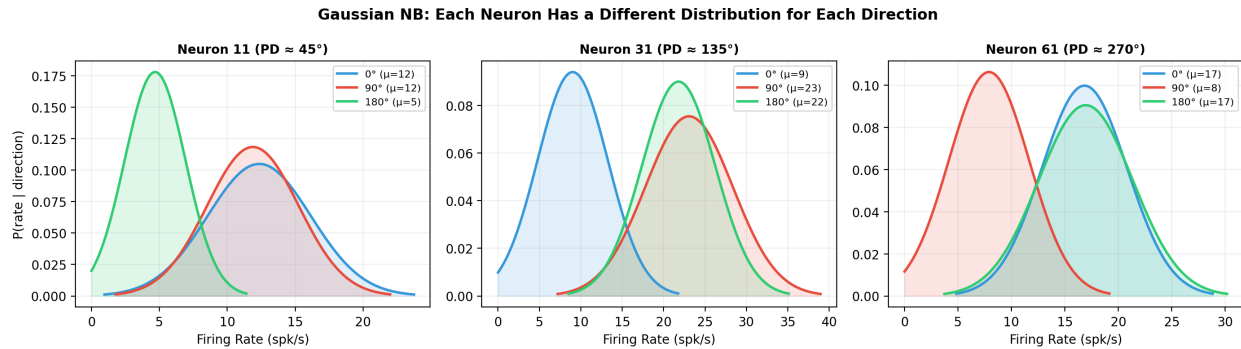


Figure 6. Class-conditional Gaussian distributions for three example neurons. Neuron 11 (PD $\approx 45^\circ$) separates 0° from 180° — its firing rate is high for one and low for the other. Neuron 31 (PD $\approx 135^\circ$) separates 90° from 0° . Neuron 61 (PD $\approx 270^\circ$) provides different information again. Because each neuron responds to a different aspect of the movement, their evidence is largely non-redundant — multiplying their likelihoods adds genuinely new information rather than double-counting the same signal.

Neuron 11 (PD $\approx 45^\circ$) and Neuron 51 (PD $\approx 225^\circ$) respond to opposite aspects of the movement — knowing one fires high tells you the other fires low, but that relationship is already captured by the class label. The independence assumption is not perfect for neurons, but it is a far better approximation than for muscles locked together in synergies. This is why NB scored 95.2% on neural direction decoding and 77.7% on neural diagnosis — each neuron's likelihood term adds roughly independent evidence, with minimal double-counting.

Why EMG direction decoding survived but binary diagnosis did not

The correlations are the same for both EMG tasks — the muscles are just as locked together whether the model is decoding direction or diagnosing impairment. The difference is in how far apart the classes are. For the 8-direction task, reaching at 0° produces a fundamentally different muscle pattern than reaching at 180° — the class means are far apart in feature space. Even with double-counted evidence, the direction with the nearest mean usually wins because the signal is so strong that the distortion does not change the ranking.

For the binary task, healthy and impaired patterns are *subtly* different — the class means are close together, and the decision boundary runs through a region where small errors in the posterior matter. Double-counting pushes NB's boundary to the wrong location: it becomes overconfident in one group when the evidence is actually ambiguous. A patient who should be classified at 55% probability of impairment might instead receive 92% — not because the evidence is strong, but because the same weak evidence was counted six times through correlated muscles.

Figure 7 makes these errors visible. Each panel plots the same 480 trials in PCA space (PC1 vs PC2), with light blue dots for healthy subjects and light red dots for impaired subjects. Trials that the model classified correctly are shown as small, faded circles. Trials that the model got wrong are shown as bold X markers: blue X = a healthy patient misclassified as impaired, red X = an impaired patient misclassified as healthy.

In panel (a), logistic regression makes relatively few errors — most clusters are clean. In panel (b), Naive Bayes is flooded with errors: 78 healthy trials are misclassified as impaired (blue X's) and 88 impaired trials are misclassified as healthy (red X's). Notice that entire subject clusters are covered in X markers — NB does not just make scattered mistakes, it systematically misclassifies some subjects on nearly every trial. These are the subjects whose muscle patterns sit in regions where the double-counted synergy evidence pushes the posterior in the wrong direction.

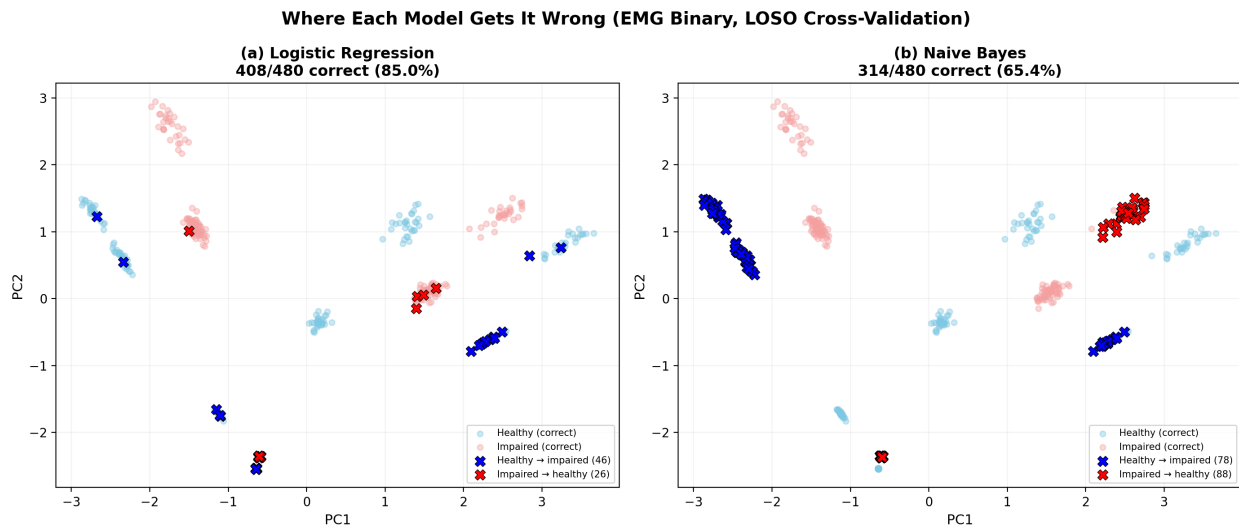


Figure 7. Misclassifications on the EMG binary task (LOSO cross-validation), projected onto PC1 vs PC2. Light circles: correctly classified trials (blue = healthy, red = impaired). Bold X markers: errors (blue X = healthy misclassified as impaired, red X = impaired misclassified as healthy). (a) LR: 85.0% accuracy, errors are sparse. (b) NB: 65.4% accuracy, errors flood entire subject clusters — 166 of 480 trials are wrong.

NB failed on EMG binary because the diagnostic signal is subtle and the features are highly correlated. The product rule double-counts the synergy signal, pushing the decision boundary to the wrong location. Direction decoding survived because the directional signal is strong enough to overwhelm the distortion. Neural features survived because their correlations are lower.

5. PCA to the Rescue — Decorrelation Fixes NB

If NB fails because features are correlated, and PCA produces uncorrelated components, then PCA + NB should fix the problem. But we must be careful about how many components to use. In Week 5, PCA with 2 components destroyed the diagnostic signal for logistic regression — the information lived in PC4, which was discarded. Now we use **all 6 PCs** — a lossless rotation that decorrelates without discarding any information.

The result is dramatic. PCA(6) + NB on the EMG binary task jumps from 65.4% to **85.0%** — a **20 percentage point gain** that exactly matches logistic regression's accuracy. The AUC improves from 0.68 to 0.91, nearly closing the gap with LR's 0.94. For direction decoding, the change is negligible (79.8% vs 81.2%).

The Fix: PCA Removes Correlations, NB Recovers

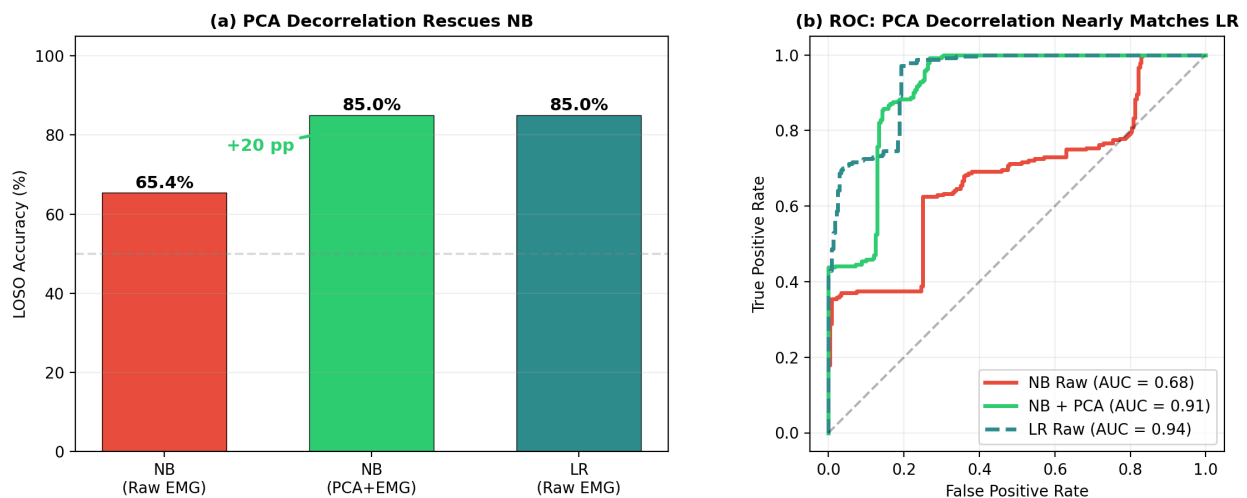


Figure 8. (a) The PCA decorrelation rescue: NB on raw EMG (65%) jumps to 85% after PCA, matching logistic regression. (b) ROC curves: NB raw (red, AUC 0.68) vs NB + PCA (green, AUC 0.91) vs LR (teal dashed, AUC 0.94). PCA decorrelation nearly closes the gap.

Why does this work? All 6 PCs is a rotation of the original 6 muscles. No information is lost — the features are merely re-expressed in an orthogonal coordinate system where NB's independence assumption is satisfied by construction. The posterior probabilities become properly calibrated, and the decision boundary moves to the correct location.

PCA's three roles across the course

This is now the third time we have used PCA, and each time it served a different purpose with a different outcome:

PCA Is Not Inherently Good or Bad — Its Value Depends on the Downstream Model

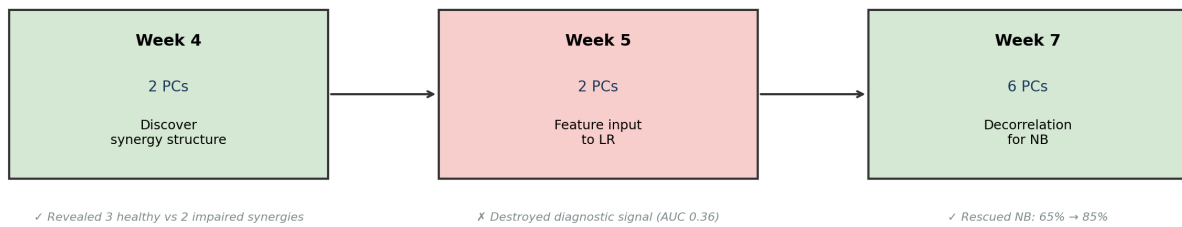


Figure 9. Three roles of PCA. Week 4: 2 PCs revealed synergy structure (useful for understanding). Week 5: 2 PCs as input to LR destroyed the diagnostic signal (harmful — information in PC4 was lost). Week 7: 6 PCs decorrelated EMG for NB, rescuing accuracy by +20 pp (essential — NB needs independent features).

PCA is not inherently good or bad — its value depends entirely on what the downstream model needs. LR's decision is a weighted sum, so correlated features cannot compound their evidence. NB's decision is a product of independent terms, so it requires uncorrelated features. Same preprocessing tool, different model needs, opposite outcomes.

6. What This Means for Motor Control Research

The NB failure on EMG is not an obscure technical issue — it reflects a fundamental property of movement data. Muscles are organized in synergies, joint angles are kinematically coupled, EMG channels share cross-talk, and force components covary during multi-joint movements. Correlated features are the *default* in motor control, not the exception. Any researcher using probabilistic models on movement data needs to ask: does my model assume independence, and if so, is that assumption defensible for my data?

Practical guideline: decompose before classifying

If your features are correlated and you want to use a generative classifier like NB, you need to decorrelate first. This week showed that PCA rescues NB's accuracy by rotating the data into uncorrelated components. But PCA has a limitation for clinical applications: its components are linear mixtures of all features, including negative weights, which makes them hard to interpret physiologically. Telling a clinician that a patient's PC3 score is abnormal is not actionable — which muscles should they target in rehabilitation?

This is where NMF (non-negative matrix factorization) from Week 4 becomes relevant again. NMF decomposes EMG into synergy modules — groups of muscles that are always activated together, with non-negative weights that map onto real muscle recruitment patterns. Unlike PCA's abstract components, NMF's synergies are physiologically interpretable: 'this patient's flexor synergy is overactive' is a statement a clinician can act on. And because NMF modules are approximately independent (each synergy responds to a different aspect of the movement), they satisfy NB's independence assumption far better than raw muscle amplitudes.

For neural data, the situation is more forgiving. Neurons with diverse preferred directions already provide approximately independent evidence, and NB performed well without any preprocessing. PCA may still help for neural populations with strong shared variability (noise correlations), but it is less critical than for EMG.

A decision framework

When choosing a classifier for motor control data, two questions matter. First: are your features correlated? For EMG, the answer is almost always yes — muscles are coupled by synergies, cross-talk, and biomechanical constraints. For neural populations with diverse tuning, correlations tend to be lower. Second: does your model assume independence? Naive Bayes does; logistic regression does not. If the answer to both questions is yes — correlated features and an independence-assuming model — you need a preprocessing step to decorrelate. For EMG, NMF is the natural choice because it produces interpretable synergy modules. For neural data, PCA is sufficient.

Correlated features are the default in motor control. If your model assumes independence, decompose first. For EMG, prefer NMF — it produces the physiologically interpretable synergy modules from Week 4. For neural data, PCA suffices, or you may not need decomposition at all.

Sidebar: What If We Binarize the Data?

Throughout this week we used **Gaussian NB**, which models each feature as a continuous Gaussian distribution. But the Naive Bayes framework is more general — the choice of likelihood function should match the data representation. **Bernoulli NB** treats each feature as binary (0 or 1) and models the probability that each feature is “on” for each class. For neural data, this means binarizing each neuron's firing rate: above baseline = 1, below = 0.

The results are instructive. For direction decoding, Bernoulli NB on binarized neural rates matches Gaussian NB almost exactly (95.6% vs 95.2%). This makes sense: cosine tuning means that whether a neuron fires above or below its baseline is highly informative about direction — the binary representation preserves the key signal. But for clinical diagnosis, Bernoulli NB collapses to 52% (near chance), while Gaussian NB achieves 77.7%. The difference between healthy and impaired is a subtle change in firing rate *amplitude* — binarizing destroys exactly this information.

The lesson: the choice of NB variant is a modeling decision about what information matters. Gaussian NB preserves amplitude. Bernoulli NB preserves presence/absence. Multinomial NB (not tested here) would preserve counts. For motor control data, where both directional tuning and amplitude modulation carry information, Gaussian NB is the natural default — but researchers working with binary event data (spike detected / not detected) might prefer Bernoulli NB for direction decoding tasks where it performs equally well with a simpler model.

7. Bringing It Together

This week we introduced our second classification algorithm and discovered that the choice of model interacts dramatically with the structure of the data. Where does Naive Bayes fit in the bigger picture?

The updated comparison table

Running Comparison Table — Through Week 7
(New rows highlighted; red = independence assumption failure)

Week	Method	Features	8-Dir Accuracy	Binary Accuracy	AUC
5	Logistic Reg (C=10)	Raw EMG (6)	80.8%	85.0%	0.94
6	Logistic Reg (C=10)	Neural (80)	91.2%	73.3%	0.80
6	Population Vector	Neural (80)	95.8%	N/A	N/A
7	Gaussian NB	Raw EMG (6)	81.2%	65.4%	0.68
7	Gaussian NB	Neural (80)	95.2%	77.7%	0.86

Figure 10. Running comparison table through Week 7. The two new NB rows (highlighted) show a split pattern: NB matches or beats LR for direction decoding but collapses on EMG binary diagnosis (65.4%, shown in red). PCA + NB rescues accuracy to match LR. All metrics use LOSO cross-validation.

Three lessons from the table

First, assumptions matter more than algorithms. NB and LR achieve nearly identical direction accuracy on EMG (81% vs 81%) and NB wins on neural (95% vs 91%), but NB collapses on EMG diagnosis (65% vs 85% for LR). The difference is not computational power — it is whether the model's assumptions match the data's correlation structure. NB assumes independence; EMG features are maximally correlated through synergies. The mismatch is fatal for subtle signals.

Second, the same preprocessing step can help or hurt depending on the model. PCA with 2 components destroyed the diagnostic signal for LR in Week 5 (AUC dropped from 0.92 to 0.36) because the information lived in PC4, which was discarded. PCA with all 6 components rescued NB in Week 7 (accuracy jumped from 65% to 85%) because it decorrelated the features without discarding any information. Same tool, different configuration, opposite outcomes — because the two models have different needs.

Third, the connection between motor control and ML runs both ways. The motor synergies from Week 4 (a neuroscience discovery) directly caused the ML failure in Week 7 (NB's collapse on correlated EMG). And the ML solution (decorrelation) points back to motor control: NMF-extracted synergy modules may be better features than raw muscle amplitudes, not just for NB, but for any analysis that benefits from independent, interpretable features.

What comes next

Week 8 introduces K-Nearest Neighbors — a non-parametric method that makes no distributional assumptions at all. It does not assume Gaussian features (like NB) or a linear boundary (like LR). Instead, it classifies by finding the most similar training examples. The question shifts: does freedom from assumptions help, or does the curse of dimensionality hit our 80-dimensional

neural features?

What we learned this week

- Naive Bayes is a generative classifier that models $P(\text{features} \mid \text{class})$ and inverts via Bayes' theorem. It gives posterior probabilities over all classes — richer output than a single label.
- The product rule multiplies likelihoods across features, assuming conditional independence. Each feature contributes to a sum of log-likelihoods (Section 2), which are exponentiated and normalized into posteriors.
- The independence assumption is catastrophic for correlated EMG features (65.4% binary) but harmless for direction decoding (81.2%) and beneficial for neural features (95.2% direction).
- PCA decorrelation rescues NB: 65% $\rightarrow$ 85% on EMG binary. For EMG, NMF may be even better — it produces physiologically interpretable synergy modules rather than abstract components.
- Correlated features are the default in motor control. When choosing a model, check whether its assumptions match your data's correlation structure. If they don't, decompose first.

Before trusting an algorithm's output, check whether its assumptions match your data. The comparison table is no longer just tracking accuracy — it is mapping which assumptions hold for which data types. That map will grow richer with every new algorithm.